ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B. Tech. (CSE)(AI)

Semester: VI

SUBJECT CODE: BCSAI601

BIG DATA ANALYTICS

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Define the following terms: (1 × 5 = 5 Marks)

(i) Define Big Data in one sentence.

(ii) List the main characteristics of big data.

(iii) What makes Big Data different from traditional data?

(iv) Which industries benefit the most from Big Data?

(v) Define HDFS in brief.

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) How is Data Science different from Data Analytics?

(ii) What is the need of MapReduce model?

(iii) What is the purpose of ETL (Extract, Transform, Load) in data warehousing?

(iv) What is the role of the Secondary NameNode?

(v) Define "support" in association rule mining.

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) Discuss the importance of data visualization in Big Data analytics. Provide examples of visualization techniques used.

OR

(b) What are the four types of analytics (descriptive, diagnostic, predictive, prescriptive)?

(ii)(a) Formulate how big data analytics helps business people to increase their revenue. Discuss with any one real time application.

OR

(b) What are some real-world applications of Big Data across industries?

Q.4.(i) (a) Demonstrate the approaches to improve Apriori efficiency.

OR

(b) What are the roles of NameNode, Secondary NameNode, and DataNode in HDFS?

(ii) (a) Explain the key differences between traditional Business Intelligence (BI) and Big Data Analytics.

OR

(b) What do you mean by Market Basket analysis and how it can help in a supermarket?

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5.(a) Explain the complete execution flow of a MapReduce job, covering all phases.

OR

(b) Explain architecture of HDFS with suitable block diagram. Discuss the role of each component in detail.

Q.6.(a) Discuss the role of ETL (Extract, Transform, Load) in Big Data processing.

OR

(b) Describe a real-world application of MapReduce, detailing its implementation.